2

Perfusion Bioreactors: The Set-Up and Process Characterisation

In this chapter, we give an overview of the challenges and objectives in operating mammalian cell perfusion cultures and provide guidelines for the design and set-up of lab-scale bioreactor systems. Next, we illustrate the control structure needed to maintain long-term stable and viable cultures and then provide a section on the media design. In the last part, we discuss steady-state operation, reactor and process dynamics, including product quality considerations. We also discuss comparisons with current technologies, particularly with respect to product quality.

2.1 CHALLENGES AND OBJECTIVES IN PERFUSION CULTURE OPERATION

Before entering the details of the realisation, control and operation of perfusion bioreactors at lab scale, it is convenient to put in perspective the challenges and the objectives connected to the application of these units.

2.1.1 Continuous Operation Mode

Depending on the specific application, continuous or perfusion bioreactors can enter in different ways in biomanufacturing. The combination of continuous nutrient supply and removal of the protein of interest and the toxic by-products coupled to cell retention inside the bioreactor allows high cell densities to be generated while maintaining the cells in a viable and growing state. In this context, we distinguish two different types of operation: on the one hand, continuous cultures without an additional cell-removing stream – the so-called bleed stream, typically applied for the generation of high cell density cultures – and on the other hand, continuous cultures with such an additional cell-removing stream, which enables long-term perfusion cultures and the achievement of steady-state conditions.

In the first case, continuous units are operated in seed trains, either for the generation of high density cell banks in perfused mode or for the operation of N-1 perfusion cultures, both targeting high cell density inoculum for either batch or continuous processes (Heidemann et al. 2010, Pohlscheidt et al. 2013, Yang et al. 2014). Another application is the intensification of batch bioreactors, the so-called concentrated fed-batch (CFB) operation. During a CFB, not only the cells are retained in the bioreactor but also the protein itself, while toxic by-products are removed in the harvest stream. This is implemented by using a filtration device with a smaller pore size – e.g., 50 kDa instead of 0.22 μm. In this case, the process is operated at a medium exchange rate enabling cell growth to much higher densities before reaching a medium limitation. The continuous operation intensifies the traditional batch-technology enabling up to a fivefold increase of cell density and harvest titre (Yang et al. 2016). Another application of continuous operation is the non-steady-state operation mode. The set-up is similar to a common perfusion process, but it is operated at a given medium exchange rate, continuously removing toxic by-products and the protein of interest. As cells are retained in the system, they grow until medium limitations are reached in terms of nutrient depletion at the given rate of medium exchange (Du et al. 2015, Gomez et al. 2017). All these systems are intended to increase the viable cell density and harvest yields compared to the traditional fed-batch processes.

The second type of operation of perfusion cultures targets long-term steady-state cultures, as in the context of the integrated continuous manufacturing of therapeutic proteins. In this case, it is not only necessary to guarantee accurate control of medium supply and perfusion rate and constant reactor volume but also to remove cells from the bioreactor in order to keep the viable cell density inside the bioreactor constant, as described in the previous chapter. To that end, cells are removed through the bleed stream, allowing the operation at constant cell density and, thus, at steady-state conditions.

It is worth pointing out that a continuous perfusion process typically consists of two different phases: an initial cell accumulation phase where sufficiently high cell numbers are generated for the next production phase, where the cell density is maintained stable at a constant level. This can be performed either in one reactor or split into an N-1 seed culture and an N-stage production reactor. During the first phase, high perfusion rates are chosen to favour cell growth, and during the production phase, perfusion is decreased and maintained at an optimal level for producing the target protein with minimum cell growth and medium consumption.

2.1.2 Challenges

In general, the operation of perfusion cultures requires solid design, robust units and a reliable control strategy of the bioreactor system. The addition of a cell retention device adds complexity and limitations to the whole system, which requires specific engineering development to deal with issues like aeration, carbon dioxide accumulation, mixing and homogenisation, fouling and failure of the cell retention device, process monitoring and control and so on. Operation at high cell densities up to 10^8 cells/mL leads, in fact, to higher overall supply and consumption rates of metabolites and gases, as well as higher viscosity values which – in combination with the shear sensitivity of mammalian cells – require specialised technological solutions.

In terms of gas supply and removal, this implies that the set-up has to support high volumetric gas–liquid mass transfer rates ($k_L a$) and volumetric gas flowrates for sufficient supply of oxygen and removal of carbon dioxide (Gray et al. 1996, Sieblist et al. 2011). For example, in order to achieve and maintain viable cell densities in the order of 10^8 cells/mL, large oxygen mass transfer coefficients in the order of 5–55 h^{-1} are necessary (Ozturk 1996), while carbon dioxide partial pressures should not exceed 100–150 mmHg (Kimura & Miller 1996, Pattison et al. 2000, Zhu et al. 2005).

A typical problem at high cell densities, particularly at larger scales, is to enable sufficient homogenisation of the cell culture broth. Insufficient homogenisation may lead to spatial gradients in both nutrient and gaseous compositions, which may affect the cellular metabolism and, thus, the achievable viable cell density. This requires a careful design of the mixing conditions – including the selection of the impeller, the baffle configuration and the stirring speed – to guarantee sufficient mixing without providing excessive stress that would harm the cell viability (Ozturk & Hu 2006).

A robust control strategy of the bioreactor conditions and culture-specific variables is important also to obtain stable perfusion cultures. This includes the tight, short-term control of the different inlet and outlet mass flowrates, viable cell density and reactor environment (pH, temperature, oxygen and carbon dioxide levels) while maintaining a constant reactor volume. Particularly relevant in these devices are long-term stability issues which relate both to mechanical devices, like preventing plugging due to fouling and protein retention in the cell retention device, as well as to the cell metabolism and the possible occurrence of ageing or stability issues (Ozturk 2014).

2.1.3 Objectives

Depending on the type of application, different requirements with respect to the bioreactor set-up, operation and control have to be considered. However, in general, the following three objectives need to be realised in order to fully exploit the potential of perfusion cultures.

The first and probably most important one is to run a stable process. This includes the mechanical and biological aspects described earlier. In the particular context of an integrated continuous process where the perfusion bioreactor is directly connected to the subsequent capture step, this becomes a prerequisite to allow stable operation of the entire downstream portion of the plant.

A second point is to achieve high productivity and medium utilisation during protein production, whether this refers to concentrated fed-batch, perfused non-steady-state culture or long-term steady-state operation. For this, the bioreactor operation has to be optimised in terms of high viable cell density, high titre and medium utilisation. In all applications, as well when generating high cell density banks and inoculum, it is important to achieve high cell-specific productivity values.

The third objective refers to the control and uniformity of product quality attributes. During steady-state perfusion cultures, the cells operate in an environment which is constant in time and, therefore, tend to maintain their metabolism unchanged and then always produce the same protein. Coupled to the constant and short product residence time within the reactor, final product quality patterns, including charge variant and glycosylation, are expected to be more uniform over time. Less exposure to protein-degrading enzymes due to shorter residence time is, in fact, expected to significantly reduce the probability of post-translational transformations such as protein fragmentation and aggregation, protein misfolding and as various types of degradation reactions – like deamidation, oxidation, sialylation, C-terminal lysine cleavage and isomerisation (Jordan & Jenkins 2007, Khawli et al. 2010). On the other hand, the presence of a continuous feed facilitates the introduction of suitable supplements in the bioreactor so as to suitably modify the product quality attributes.

The ability to properly design and develop perfusion bioreactors so as to deal with these objectives is the key to exploit the full potential of these units.

2.2 EQUIPMENT

Let us now consider the equipment necessary to develop and operate a lab-scale bioreactor system that meets the requirements for a stable perfusion culture. Besides the bioreactor system itself, equipped with proper control devices, this includes sensors for monitoring culture-specific parameters and suitable cell retention devices.

2.2.1 Bioreactor

Perfusion cultures can be operated in various different devices ranging from single-use units, like wavebag bioreactors (Clincke et al. 2013a, 2013b), to the classical stirred tank vessels (Karst et al. 2016, Xu & Chen 2016). For the purpose of running a continuous culture, the reactor system should be equipped with a number of connecting ports so as to accommodate all necessary tubes and fittings.

Whether operated in batch or continuous mode, bioreactors require reliable monitoring and control of various operating variables such as temperature, pH and dissolved oxygen (DO), through the proper operation of various pumps and mixing devices. In addition, the perfusion reactor system should enable the connection to medium, harvest and bleed bags or holding vessels, as well as the connection to a cell retention device that typically represents an additional part of the set-up. This can either be an external device or implemented inside the bioreactor itself (Zhang et al. 2015b). Besides the control units, the reactor has to be equipped with a stirring/mixing device, an aeration/sparger module, a condenser, sample and inoculation ports, connections for the addition of antifoam and base for pH control. Devices for aseptic connection are also necessary – e.g., a sterile welder and sealers.

For the configuration of the bioreactor, suitable geometrical parameters have to be chosen. In the case of a stirred vessel, these are particularly relevant at the large scale. Such parameters include the type of stirrers and baffles and all the relevant distances, such as the ratio of impeller and tank diameter or the impeller distance from the reactor bottom. A suitable sparger and stirrer that allow breakage and complete dispersion of gas bubbles at reasonable impeller speeds and elevated gas flowrates must also be chosen, without harming the cell viability. These aspects are discussed in Chapter 5, where suitable design criteria are also provided.

2.2.2 Sensors

The proper monitoring and control of the bioreactor operation necessitates several sensors. These include a pH probe, DO probe, temperature sensor, CO_2 sensor and biomass probe. The pH, DO and temperature probe represent industrial standards, often part of the bioreactor system modules, which are used to monitor and control the operation at predefined operating set points. Monitoring and control of CO_2 are important because high levels of dissolved CO_2 result in reduced cellular growth and productivity (Garnier et al. 1996, Kimura & Miller 1996). Biomass probes, usually based on capacitance measurements, are used for the online monitoring and control of the biomass inside the bioreactor that is often correlated to the viable cell density. In particular, biocapacitance has been shown to be linearly correlated to the viable cell density and, therefore, provides the ideal tool for monitoring and controlling this important process parameter (Carvell & Dowd 2006).

In addition, more complex sensors such as Raman, near infrared (NIR) or fluorescence sensors are and will be increasingly considered. These typically need to be coupled with suitable statistical multivariable data analysis techniques in order to estimate, from the measured complex spectra, process variables such as metabolite and protein concentrations or critical quality attributes, like N-linked glycosylation patterns or aggregates (Luttmann et al. 2012, Mercier et al. 2016, Rathore 2014, Rathore et al. 2018). However, applications are currently still limited to batch operation modes. The extension to longer operation times requires robust sensors with long lifetime, resistant to fouling and sterilisation procedures (e.g., autoclaving, gamma sterilisation). A more complete discussion of these and other monitoring devices is provided in Chapter 3.

2.2.3 Cell Retention Device

The cell retention device is a key element of perfusion cultures, and its longevity and scalability are fundamental for the long-term stability and reliability of the process. Cell retention systems are based on different physical methods for cell separation – in most cases, either size or density. The immobilisation and entrapment of cells in larger beads or fibres could, in principle, provide an effective method for retaining cells inside a bioreactor, but their operation and handling are often not convenient (Ozturk & Kompala 2006). In addition, there are also strong limitations in terms of maximum viable cell density, dictated by the limited availability of surface area compared to volume, like in suspension cultures. More suitable cell retention devices include cross-flow filtration systems, hollow-fibre modules, spin filters, gravity settlers, centrifuges and acoustic wave separators (Bielser et al. 2018, Bonham-Carter & Shevitz 2011, Voisard et al. 2003). Each of these shows advantages and drawbacks. Retention by size is typically used in filtration devices that use a physical barrier to separate cells, both viable and dead, from the cell culture medium. This allows high separation efficiencies. The major drawback of separation by size is that these systems tend to foul and clog. In the case of retention by density, the separation relies on the density difference between cells and culture medium without requiring filtering devices, thus making it more robust and suited for long-term operation. However, retention by density often comes along with long residence times of the cells within the retention device due to the rather small density difference between cells and medium, leading to low gravitational settling velocities (1–15 cm/h) (Voisard et al. 2003).

The time the cells spend in the retention device is, in fact, an important variable which should be kept rather short. The environment in these units is not controlled and differs from the one inside the bioreactor. Therefore, it might irreversibly impact cellular performance, as a drop in temperature, oxygen and nutrient limitations, or lower pH levels can affect cell metabolism.

On the whole, it is seen that a reliable cell retention device has to fulfil several requirements. These include ideally full cell retention, prevention of product sieving, operation at feasible perfusion rates, sufficient oxygen supply, prevention

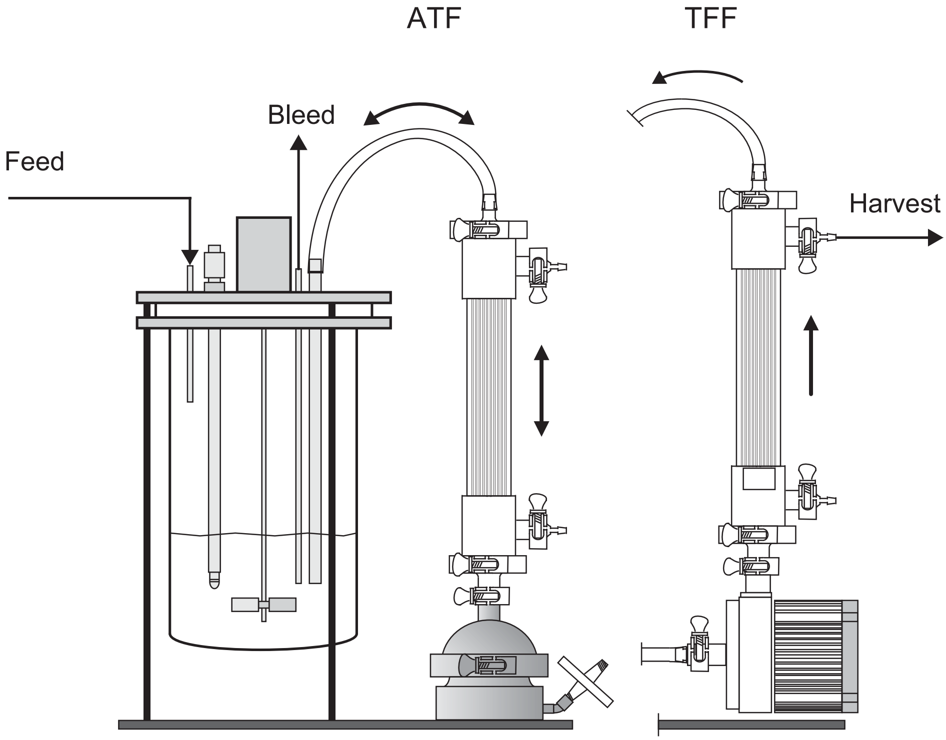

Figure 2.1 Schematic representation of the set-ups used for the operation of the ATF (left) and TFF (right) retention devices in continuous manufacturing. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

of cell harmful shear stress, relatively short residence time and long-term operation without failure.

Today, tangential filtration devices such as the alternating tangential flow (ATF) and the tangential flow filtration (TFF) hollow-fibre systems represent industrial standard technologies and are typically preferred over the other ones. In the case of the ATF system, a diaphragm pump is connected to a hollow-fibre module (typical pore size: 0.22 μm) that is directly connected to the bioreactor system. The pump uses alternating back and forward pushes to exchange the culture broth between the reactor and the hollow-fibre module. In the case of the TFF system, a pump is directly connected to the bioreactor outlet and to the hollow fibre and moves the cell culture broth into an external loop, through the hollow fibre and back to the bioreactor system. The two operations are illustrated in Figure 2.1. In both systems, high liquid flowrates at the membrane surface can reduce filter fouling and keep the operation stable. In particular, it has been shown that the alternating flow within the ATF device leads to a specific beneficial self-cleaning effect of the fibre module (Bonham-Carter & Shevitz 2011). Furthermore, these technologies have advanced in recent years and nowadays commercial solutions exist for lab to manufacturing scale (1–1,000 L) consisting either of a single-use or a stainless steel device.

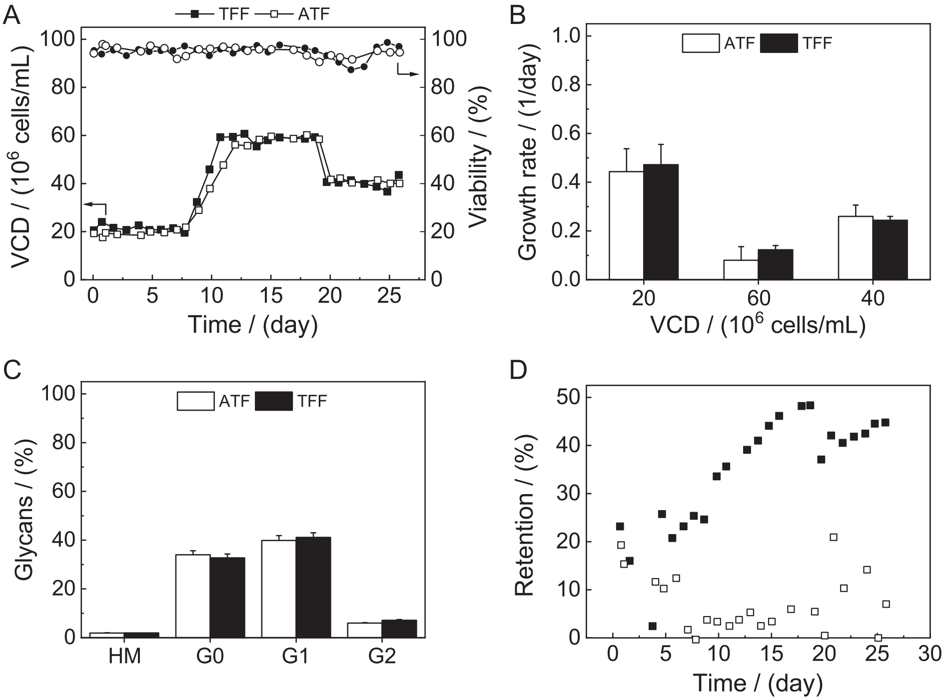

Figure 2.2 Comparative studies of the ATF and the TFF device on benchtop bioreactors using the same hollow fibre at three different viable cell density set points: (A) Viable cell density and viability, (B) Specific growth rates, (C) N-Glycan distribution and (D) product retention. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

Comparative studies of the ATF and the TFF devices using the same hollow-fibre have been reported by Karst et al. (2016). A laboratory bioreactor equipped with either the TFF or the ATF device was operated through three consecutive steady states, with VCD set points equal to 20, 60 and 40 $\times$ 10^6 cells/mL, respectively, as shown in Figure 2.2A. While process performance (cellular growth and metabolic input and output rates) as well as product quality of the produced mAb were similar between the two systems – as illustrated in Figure 2.2B and C, respectively – Figure 2.2D shows that a significantly larger portion of the protein was retained in the bioreactor equipped with the TFF device.

In a different study, Wang et al. (2017) also compared the process performance of the ATF and TFF devices. In the first step, they replaced a peristaltic pump by a low-shear centrifugal pump to operate the TFF, which dramatically reduced product sieving and led to comparable levels of sieving between the ATF and TFF systems. Karst et al. (2016) and Wang et al. (2017) used a similar type of centrifugal pump for their comparative studies but apparently obtained different results. While Karst et al. (2016) observed higher product retention in the TFF device, Wang et al. (2017) obtained similar levels of product retention in the ATF and TFF devices. This is, in fact, a system-dependent behaviour, so considering different cell lines, medium compositions and protein products may lead to different observations.

Using ATF and TFF retention devices in connection to wavebag bioreactors revealed that very high cell densities are limited by the vacuum capacity of the ATF, thus resulting in a maximum viable cell density of 1.32×10^8 cells/mL, compared to 2.14×10^8 cells/mL in the TFF device (Clincke et al. 2013a). This upper bound was imposed in the TFF by the high pressure inside the TFF loop due to the increased viscosity of the cell broth at such high flowrates. Additionally, too high pCO_2 levels and insufficient oxygenation were observed. On the other hand, comparing the ATF device to an internal spin filter system, Bosco et al. (2017) observed that the ATF system supports higher maximum perfusion rates and, consequently, higher maximum cell densities.

In acoustic filter systems, cell-free harvest is enabled by ultrasonic separation of cells and medium. The exposure to an acoustic resonance field leads to an increase in the biomass particulate size as a result of three different radiation forces inducing cellular aggregation. Separation efficiencies of greater than 90 per cent were achieved for cell densities below 10×10^6 cells/mL and high perfusion rates (Baptista et al. 2013, Gorenflo et al. 2003). However, this technique shows a narrow range of allowed operating conditions for optimal cell retention the cell separation efficiency is limited by cell concentration and input power, and process scale-up is difficult (Pörtner 2015).

This variety of studies shows the importance of cell retention in the development and design of perfusion bioreactors and highlights ATF and TFF systems as the current state-of-the-art technology at lab and manufacturing scale. Note that, just like sensors, the cell retention devices need to be resistant to sterilisation procedures (e.g., autoclaving and gamma sterilisation).

2.2.4 Configuration

Different lab-scale systems for the development of perfusion processes have been described in the literature. One example that we consider as being a good reference is the system developed by Karst et al. (2016). This consists of a stirred tank reactor (2.5 L glass vessel), typically operated at 1.5 L and equipped with a single six-blade Rushton turbine (RT) impeller that allows breakage and complete dispersion of gas bubbles at reasonable impeller speeds (200–400 rpm). Together with an open pipe sparger, this system allows both high oxygen mass transfer coefficients and high volumetric gas flowrates. The stirred tank reactor was coupled with either an ATF or a TFF cell retention device and then placed on a balance to implement weight-based flowrate control. A suitable head plate configuration in such a set-up includes the connection of the bioreactor to the cell retention device, all sensors and sufficient ports for the different inlet and outlet streams. The bioreactor set-up and head plate configuration are illustrated in Figure 2.3.

The developed system showed high mixing effectiveness with low shear and suitable gas–liquid mass transfers, with k_La values in the range 5–40 h^{-1}, sufficient to support up to 10^8 cells/mL (Ozturk 1996). A more detailed characterisation of reactor set-ups in terms of hydrodynamic stress, gas–liquid mass transfer, and mixing effectiveness, particularly at larger scales, is discussed in Chapter 5.

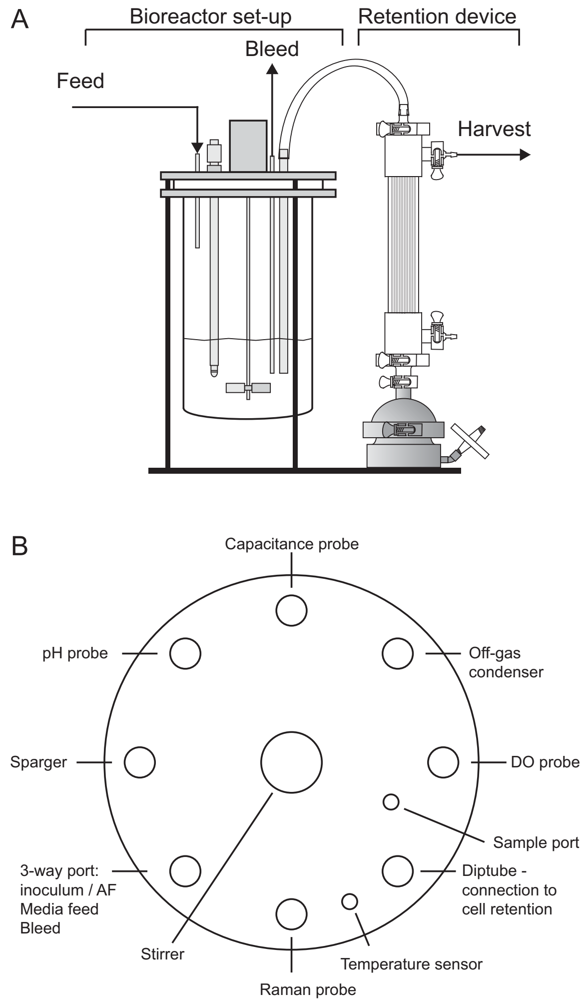

Figure 2.3 Overview of the bioreactor set-up (A) side view and (B) schematic of the head plate configuration.
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2.3 PERFUSION-SPECIFIC CONTROL VARIABLES: P, VCD AND CSPR

In this section, the most important parameters that characterise perfusion processes and that are necessary for process control are discussed. In these processes, the flowrates are commonly expressed as specific rates – that is, volume of medium per volume of bioreactor per day (VVD) or simply reactor volume (RV) per day. Consequently, the perfusion rate is the ratio of the volumetric feed flowrate, Q_{feed}, and the total reactor volume, V_R:

$$P = \frac{Q_{feed}}{V_R}. \tag{2.1}$$

In order to keep the reactor volume constant, the sum of the rate of cell-free supernatant removal – the harvest rate, H – and the rate of cell discard – the bleed rate, B – need to equal the perfusion rate:

$$P = H + B. \tag{2.2}$$

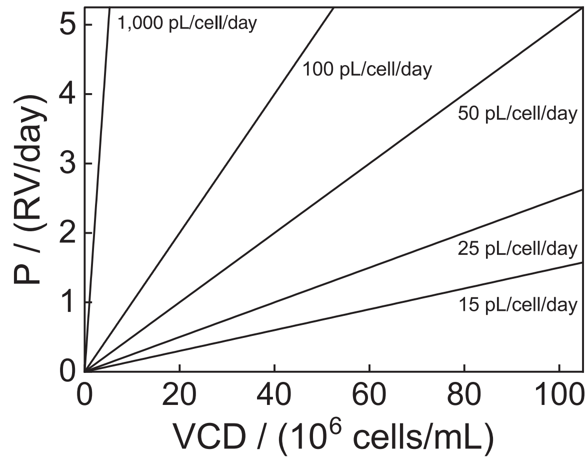

Figure 2.4 The CSPR describes the ratio of the perfusion rate (P) and the viable cell density (VCD). The lower the CSPR, the flatter is the slope of the straight line and, thus, the less medium is provided per cell.

A very significant operating parameter in perfusion processes is the cell-specific perfusion rate (CSPR), which relates the perfusion rate and the viable cell density (VCD), X_V and expresses the amount of medium fed to a single cell per day:

$$\mathrm{CSPR} = \frac{P}{X_V}. \tag{2.3}$$

The CSPR is generally expressed in pL/cell/day or nL/cell/day. In Figure 2.4, the relation between perfusion rate and viable cell density is illustrated. It is seen that the lower the value of the CSPR, the lower the perfusion rate is to maintain a certain viable cell density. For example, in the case we want to operate at a viable cell density of 20×10^6 cells/mL with a CSPR of 100 pL/cell/day, the perfusion rate has to be set at 2 RV/day. When we choose the CSPR to be 50 pL/cell/day for the same VCD, the perfusion rate has to be set to 1 RV/day. A further decrease to 25 pL/cell/day would require a decrease of the perfusion rate to 0.5 RV/day.

It is clear that in order to sustain a certain VCD value, it is necessary to provide sufficient nutrients. This implies that there is a minimum value of the CSPR, below which the operation becomes unstable and cells start to die. This is why the CSPR represents a key quantity in the design, optimisation and control of perfusion bioreactors, and it should be kept above such a minimum value, often referred to as minimum cell-specific perfusion rate ($\mathrm{CSPR}_{\mathrm{min}}$). Accordingly, this provides a direct performance indicator for the given medium and expression system: the so-called medium depth. By maintaining such a CSPR value constant, cell metabolism and metabolite concentrations should remain constant, thus leading to a stable culture behaviour (Konstantinov et al. 2006, Ozturk 1996).

Reported minimum cell-specific perfusion rates range from 15 to 500 pL/cell/day (Dowd et al. 2003, Karst et al. 2016, Konstantinov et al. 2006, Warikoo et al. 2012, Xu & Chen 2016). Recent operations at very low CSPRs, about 15–20 pL/cell/day, indicate the improvement in medium formulations in recent years (Karst et al. 2017a, Xu & Chen 2016). The historical development of feasible CSPRs is illustrated in Figure 2.5.

It is worth pointing out that the previously mentioned operating variables need to respect very strict boundaries. The perfusion rate is lower bounded by the maximum allowable residence time in the reactor – and, therefore, by the protein stability – and

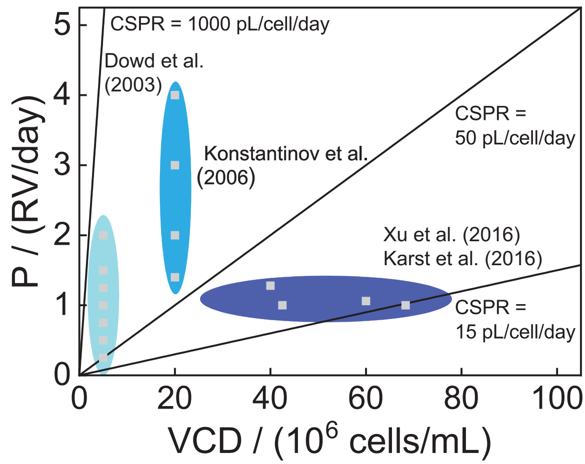

Figure 2.5 Historical development of CSPR operating ranges. In the early 2000s, Dowd et al. (2003) tested CSPRs in the range 50–400 pL/cell/day at 5×10^6 cells/mL (light blue), while Konstantinov et al. (2006) operated at CSPRs in the order of 50–400 pL/cell/day at 20×10^6 cells/mL (blue). In most recent studies, Karst et al. (2016) and Xu & Chen (2016) performed perfusion studies at CSPR values below 20 pL/cell/day (dark blue).

upper bounded by the volumetric capacity of the cell retention device. On the other hand, the maximum allowable VCD is directly linked to oxygen supply so that, in most cases, the maximum viable cell density (VCD_{max}) is limited by the maximum possible oxygen transfer rate (OTR) provided by the set-up. In addition, the overall biomass content in the bioreactor is upper bounded by the viscosity of the suspension, which affects mixing and all mass transfer processes.

2.4 CONTROL

High cell densities and elevated metabolic demands make continuous perfusion processes highly dynamic, and a robust process control is inevitably important. For a stable operation of the perfusion culture, the different flowrates (feed, harvest and bleed), the working volume and the cell density have to be tightly controlled. Special attention is given to the cell-specific perfusion rate independent of transient or steady-state operation, since this determines the stability of the culture. Altogether, several control strategies have to be combined – including classical control loops of pH, temperature, aeration and mixing – with perfusion-specific ones needed to keep inlet and outlet flows, as well as the liquid level in the reactor, constant.

2.4.1 Cell-Specific Perfusion Rate

As previously mentioned, the CSPR defines the amount of nutrient delivered to each cell per unit time. As such, it is a very important parameter for the culture operation. For example, at the (N-1) seed stage, high CSPR values are used to promote cellular growth at the expenses of an increased medium consumption. On the other hand, at the N production stage, a lower CSPR reduces nutrient delivery to the cells, leading to not only decreased cellular growth rates but also higher titres and more efficient medium utilisation.

The proper design and development of a mammalian cell perfusion process requires a careful analysis of suitable viable cell densities, perfusion rates and cell-specific perfusion rates for a given set-up, expression system and medium. Indeed, maintaining the desired CSPR during operation is needed to have a stable and efficient process. Nevertheless, this does not mean that we need to implement a control loop having CSPR as the set point. In the following section, we describe control strategies based on the control of P and VCD, with set points which obviously satisfy the desired CSPR value (Konstantinov et al. 2006), although alternative strategies are also possible (Konstantinov et al. 1996, Ozturk 1996).

2.4.2 Primary Control Loops

The bioreactor, independently of the batch or continuous operating mode, has to be operated at a fixed set of operating conditions including DO concentration, pH, temperature and stirring. Consequently, temperature control; mass flow control for air, nitrogen and oxygen to maintain the desired DO; and a tight control of the stirring speed have to be installed since the failure of any of these might have an irreversible impact on the culture. In general, similar systems to those currently used for fed-batch bioreactors can serve this purpose also in the case of perfusion cultures.

Typically, cell cultures should be operated at pH 6.9–7.2 (Miller et al. 1988, Wong et al. 1992). The control of the cultures pH is often implemented using a buffer system with CO_2 addition to decrease pH and base addition to increase pH. However, it has been demonstrated that the understanding of the interactions between the partial carbon dioxide pressure, pCO_2; lactate, a by-product of the cells; and pH can enable the operation of stirred-tank bioreactors without external pH control, thus preventing the addition of excessive amounts of base (Xu & Chen 2016). A different approach is to use only a one-sided pH control strategy by controlling the CO_2 fraction in the inlet gas, thus again preventing excessive base addition (Karst et al. 2016).

The dissolved oxygen is controlled by the fraction of oxygen (oxygen partial pressure, pO_2) and the volumetric flowrate of the inlet gaseous stream. However, too-high oxygen partial pressures can stimulate the generation of radical oxydative species (ROS) and induce cell death. Consequently, sparging of pure oxygen should be prevented (Godoy Silva et al. 2010). Commonly used process set points include a DO of 50 per cent, temperature at 36–37°C and stirring rates in the range 200–400 rpm.

2.4.3 Control of Working Volume and Perfusion Rate

The stability and robustness of perfusion cultures inevitably depends on the tight and robust control of the working volume and the perfusion rate. These two parameters are directly related. For working volume control, either the reactor has to be placed on a balance or a sensor that monitors the liquid level inside the reactor has to be used. Two control strategies can be adopted: the perfusion rate is fixed, and the harvest rate is adjusted by a proper controller so as to keep the measured weight or liquid level in the reactor constant (Figure 2.6) or, vice versa, the harvest rate is maintained constant while the feed rate is properly adjusted by the controller (Figure 2.7). Adjustment of either the perfusion/feed or the harvest rate according to the reactor weight can be regulated by a standard proportional integral derivative (PID) controller.

The choice of one or the other strategy depends on the user's application and priorities. During the process-development phase, it could be recommended to keep constant the perfusion rate. Once the process is fixed, all expected flowrates are known, and this issue becomes less important. On the other hand, in the case where the bioreactor is directly coupled to the first step of the downstream purification train – e.g., the capture step, as in the context of continuous integrated

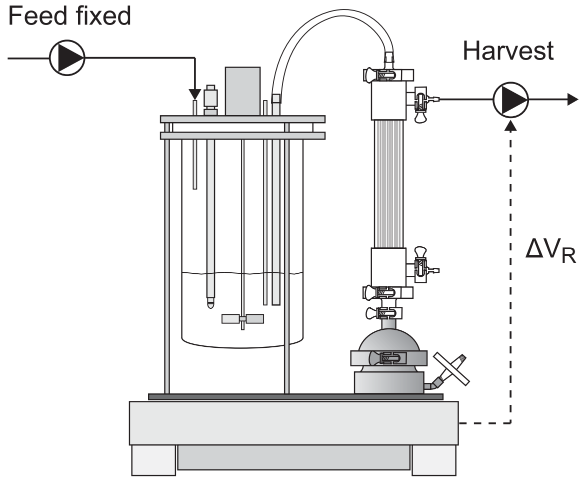

Figure 2.6 Control of working volume and harvest rate: operation at fixed perfusion/feed rate with proper adjustment of the harvest rate to control the reactor weight. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

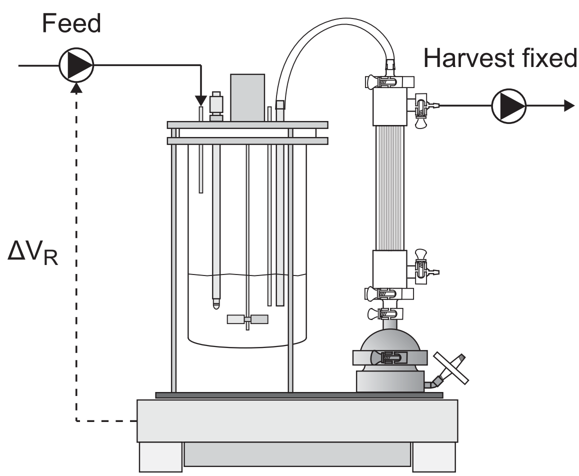

Figure 2.7 Control of working volume and feed/perfusion rate: operation at fixed harvest rate with proper adjustment of the feed/perfusion rate to control the reactor weight. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

manufacturing – it is probably better to keep constant the harvest rate, which constitutes the feed flowrate to the capture unit so as to avoid propagating possible disturbances along the downstream units.

The cell containing bleed stream is necessary to compensate for cell growth and achieve steady state. Since the bleed stream contains also the target protein, which is wasted, it is directly linked to product loss and should be kept as small as possible to maximise the process yield. The bleed rate is usually not directly included in the weight control strategy of the bioreactor as it constitutes 10 to 20 per cent of the overall perfusion rate (Deschênes et al. 2006, Lin et al. 2017). Its control is best implemented through a biomass control strategy, as described in the next section.

2.4.4 VCD/Biomass Control

The viable cell density in perfusion bioreactors is typically controlled by the bleed rate. For this, various methods for monitoring the viable cell density are used – ranging from acoustic resonance densitometry, conductivity, optical sensors and

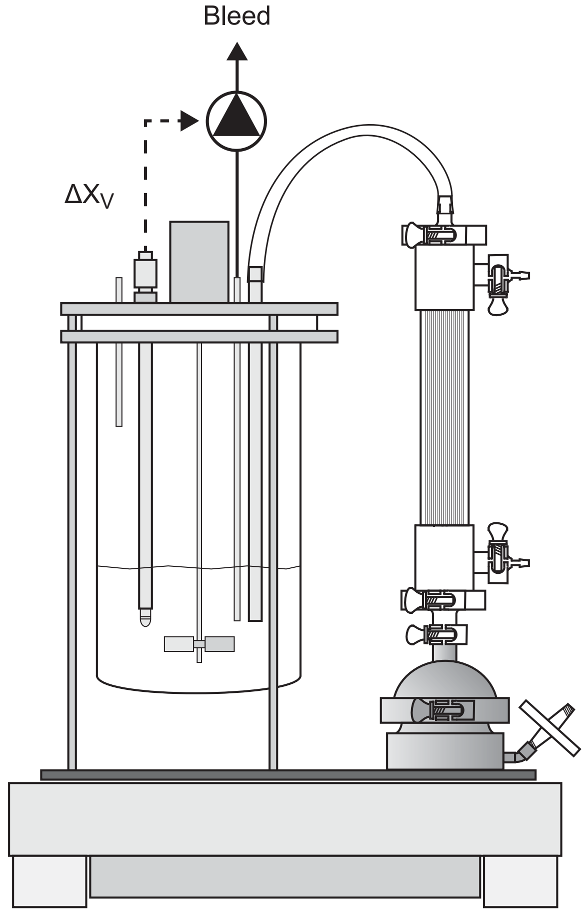

Figure 2.8 VCD/Biomass control strategy through the bleed rate. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

real-time imaging to indirect methods such as the measurement of the oxygen uptake rate or the adenosine triphosphate (ATP) production rate (Carvell & Dowd 2006, Deshpande & Heinzle 2004, Dowd et al. 2003, Kiviharju et al. 2008, Konstantinov et al. 1994, Yoon & Konstantinov 1994). The current state-of-the-art technology is based on capacitance sensors for the real-time monitoring of viable cell density.

In order to operate at a constant viable cell density, two approaches are possible. The simpler one involves a semi-continuous bleeding strategy, which is applied when no online measurement of the cell density is available. In this case, a proper portion of the bioreactor cell broth is removed once or twice per day according to measured viable cell densities, as discussed by Clincke et al. (2013a) and Karst et al. (2016). For the second approach, online monitoring of the biomass or viable cell density is necessary. The signal of the biomass probe can be used to implement an automated bleeding strategy (Deschênes et al. 2006, Konstantinov et al. 1994, Steinebach et al. 2017) as schematically indicated in Figure 2.8, where the bleed rate is adjusted through a standard PID controller to keep the biomass constant at the set point value.

It is worth noting that the system responds differently to changes in the bleed rate, depending on whether the perfusion or the harvest rate is kept constant – i.e., Figures 2.6 and 2.7, respectively. For example, in the first case, an increase in the bleed rate leads to a decreased harvest rate. On the other hand, if the harvest rate is set

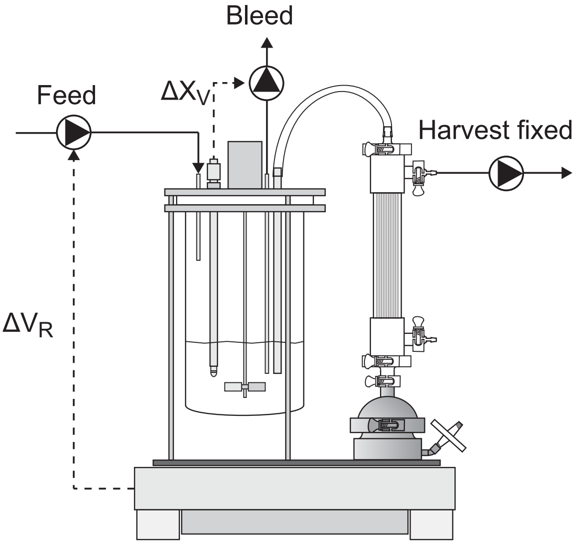

Figure 2.9 Overall bioreactor control strategy. The harvest rate is fixed and the bleed pump is controlled through a proper VCD sensor. The feed pump is adjusted to keep constant the reactor weight. © Reprinted from *Biochemical Engineering*, 110, Daniel J. Karst, Elisa Serra, Thomas K. Villiger, Miroslav Soos and Massimo Morbidelli, 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', 2016, with permission from Elsevier

constant, as the bleed increases, the perfusion/feed rate also increases, as illustrated in Figure 2.9.

2.4.5 Cell Retention Device Operation

The reliable operation of the cell retention device is a strict requirement for long-term stable perfusion cultures. For this, close monitoring and control of the exchange rate between the bioreactor and the external retention device is necessary so as to provide early indication of operational failure. In the case of the alternating tangential flow module, a diaphragm pump typically is used. This pump is controlled by a separate controller supplied by the manufacturer. The periodic alternation of pressurising and depressurising cycles results in an oscillating change of the flow direction and rate. On the other hand, in the TFF device, where the cell culture broth is pushed by a centrifugal pump through an external loop containing the retention device, it is important to carefully control the rotational speed of the pump. High rotational speed can lead to shear stresses that exceed the cells maximum tolerable stress (Karst et al. 2016, Neunstoecklin et al. 2015).

An important aspect refers to the monitoring of the degree of fouling of the hollow-fibre membrane. For this, additional pressure sensors can be mounted on the permeate and the retentate side of the membrane to measure the transmembrane pressure. An increasing transmembrane pressure and decreasing permeate flow typically indicate progressive filter fouling (Kelly et al. 2014). At lab scale (1–10 L bioreactors), exchange rates of 1.0–1.5 L/min are typically reported (Clincke et al. 2013a, Karst et al. 2016).

2.5 MEDIA FOR PERFUSION CELL CULTURES

Medium formulation plays an important role in the development and optimisation of any cell culture process. In the early days of cell culture, media consisted of natural biological substances like plasma, serum and other animal or plant extracts, which naturally contain the nutrients and growth factors necessary for the cells to grow. The use of these ingredients, however, posed several problems, including the lack of control over their composition, resulting in significant lot-to-lot variations and, consequently, unreliable cell culture processes. In addition, due to their animal origin, these components could carry pathogens like viruses, thus making these media too risky for the production of active species that would eventually be injected into a patient (Grillberger et al. 2009, Yao & Asayama 2017).

Modern cell culture media are mostly chemically defined (CD); that is, all the components are well-defined chemical species dosed at a precise concentration (Yao & Asayama 2017). This includes, for example, salts, vitamins and amino acids but also, in certain cases, recombinant proteins (highly purified and produced in a controlled environment). CD media include up to hundreds of different ingredients, and their formulation for a given culture involves the definition of all the relative concentrations. This is indeed a complex task which requires specific knowledge on cell biology (Lin et al. 2017, Ling 2015, Ozturk & Hu 2006, Reinhart et al. 2015, Yao & Asayama 2017).

The existing knowledge about media design and optimisation for fed-batch processes can be leveraged for the use in perfusion processes (Lin et al. 2017). Fed-batch media are optimised neither for the cellular needs in a perfusion process nor for some operational constraints related to the large volumes that need to be prepared, sterilised and stored before being added to the bioreactor. This is a particularly relevant point, as a proper medium design has a significant impact on process operation and constitutes a significant, if not the largest, portion of the process operating cost (Pollock et al. 2013, Xu et al. 2017b).

In the following, we discuss a few aspects specifically related to media application in perfusion, with reference to the medium depth and the medium concentration, which are both related to maximising the reactor volumetric productivity by using more efficient media. The medium depth defines the minimum perfusion rate of that medium needed to sustain a given viable cell density and corresponds to the reciprocal of the medium-specific $\mathrm{CSPR}_{\mathrm{min}}$ value (Konstantinov et al. 2006):

$$\text{Medium depth} = \frac{1}{\mathrm{CSPR}_{\mathrm{min}}} \tag{2.4}$$

For a given medium formulation – that is, for a given medium depth or $\mathrm{CSPR}_{\mathrm{min}}$ – the sustainable VCD in a perfusion bioreactor can be increased by operating at higher perfusion rates in order to provide the same amount of nutrients per cell. This corresponds to moving along the iso-CSPR lines shown in Figure 2.4, with the one corresponding to the $\mathrm{CSPR}_{\mathrm{min}}$ obviously providing the most convenient condition – that is, the largest possible viable cell density for the given perfusion rate. Improving the medium depth targets to decrease the $\mathrm{CSPR}_{\mathrm{min}}$ – and, therefore, to increase the viable cell density for a given perfusion rate – which leads to a higher volumetric productivity of the reactor.

Chemically defined media offer the possibility of taking a systematic approach to media design by investigating the effect of each individual species on the medium depth. The various ingredients can be grouped into different categories that include carbon sources, amino acids, vitamins, trace elements, salts, lipids, polyamines and hormones. Some of these are consumed by the cells as an energy source or as building blocks for housekeeping activities, cellular division and, of course, for the synthesis of the targeted recombinant protein. Other functions do not imply the component depletion, like shear stress protectants or salts containing sodium or potassium,

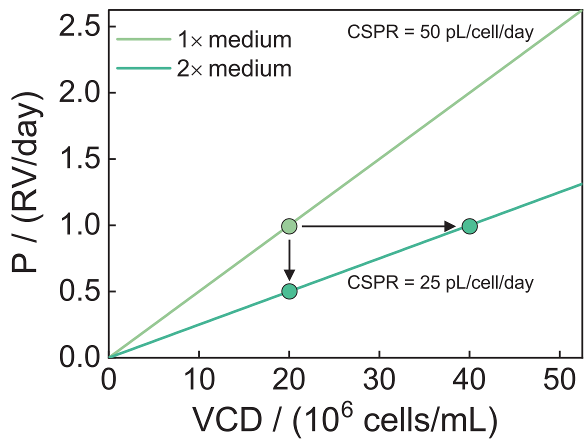

Figure 2.10 Relation of CSPR, P and VCD for a given cell line and two different media concentrations: one (2× medium) being the double of the other (1× medium).

which act on osmolality and allow the cells to build up the natural sodium potassium transmembrane gradient. Since each component can be quantified individually, it is possible to monitor its consumption during the cell culture and consequently rebalance the medium by adding some concentrated feed formulations containing all or parts of the media components. This allows grouping media components into two main categories:

- Nutrients: amino acids, vitamins, carbon sources
- Baseline components: pluronic, salts, buffers

Indeed, the selection of the hundreds of species present in a medium and of their relative composition is a difficult task since, in principle, each of them can affect the medium depth. However, since we defined the CSPR as the volume of media per cell per unit time, the medium concentration also affects the CSPR. That is, by concentrating all nutrients in the medium, we can feed to the reactor the same mass of each nutrient but with a lower volumetric perfusion flowrate, thus effectively reducing the $CSPR_{min}$ or increasing the medium depth. Medium concentration is, therefore, also an important parameter in medium design. For example, assuming that there are no physical (i.e., solubility) limitations in doubling the concentration of all medium compounds, going from a 1× medium to a 2× medium, the CSPR decreases by a factor of 2, e.g., from 50 to 25 pL/cell/day, as shown in Figure 2.10. As a result, the same cell density can be sustained by half the perfusion rate, or, alternatively, the VCD can be doubled while keeping the perfusion rate unchanged – all of this, of course, under the assumption that physiological conditions are maintained and that no other limitation, such as oxygen consumption or accumulation of toxic metabolites, arises.

However, in practice, increasing media concentration is not straightforward as several limitations must be accounted for, with solubility and stability being the primary concern. Physiological conditions imposed by the biology of the cells also include pH and osmolality constraints that the bioreactor environment cannot violate. In general, this is achieved by buffering the pH using bicarbonate and adjusting the osmolality using salts. As an example, in Figure 2.11, it is shown how to double all components in the medium (glucose, vitamins, amino acids and others), while adapting sodium chloride to keep osmolality unchanged. This

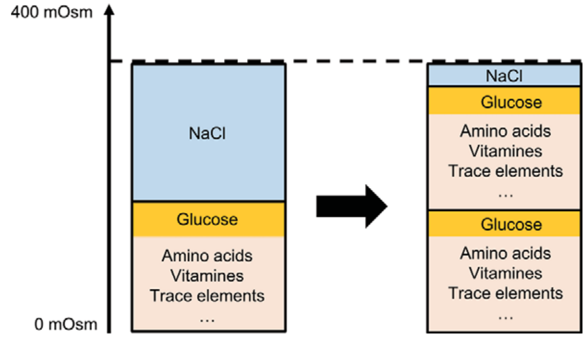

Figure 2.11 Schematic of media composition expressed in osmolality contribution: the concentration of sodium chloride is adapted to maintain osmolality unchanged while doubling the nutrient concentrations.

example is simplified and does not consider, for example, the adaptation of the pH buffer that would, in practice, change because of the increased proportion of acidic compounds.

Thus, when designing a medium for a given culture, we need to consider the function of each of its components. Metabolic needs can be understood by comparing the composition of the spent medium in the harvest stream to that of the fresh medium. From this, cell-specific consumption and production rates of the different nutrients can be computed to estimate the relative need of each nutrient. The chemical species that buffer pH, control osmolality or act as shear protectant do not necessarily need to be concentrated in order to improve the medium depth. Nevertheless, their concentration should also be adapted and optimised in the medium according to their function. For example, if the amino acids, glucose and vitamins are concentrated, the salt concentrations should be reduced to keep the osmolality constant. On the other hand, the bicarbonate level should be adapted to the new formulation to maintain its buffering action unaltered. The concentration of the shear protectant should be maintained roughly constant, usually between 0.5 and 2 g/L, although higher concentrations up to 5 g/L have been reported to be beneficial, especially at higher cell densities (Meier et al. 1999, Xu et al. 2017c). Of course, these considerations are of very general value but provide the basis for fine-tuning the concentrations of the different ingredients in the medium and improving the volumetric productivity or the footprint of a given cell culture.

In an industrial manufacturing environment, media are prepared starting from a homogeneous powder formulation, which typically represents the company's platform medium and contains most, if not all, of the compounds. This leaves no room for flexibility, for example, to concentrate the nutrients without simultaneously concentrating also the baseline components. On the other hand, in a process development set-up, it is more convenient not to include baseline components in the powder formulation to leave open the possibility of concentrating the nutrients while independently changing or keeping constant the baseline components. By adding a certain portion of such a depleted powder to a defined volume of liquid, a 1× concentrated medium is obtained. To concentrate it twice, two parts of the same powder should be added. Baseline compounds are not part of this powder formulation, and therefore, their concentration can be adjusted independently, depending of their specific function, as schematically illustrated in Figure 2.12. This implies controlling osmolality with the salts, pH with the buffer and other characteristics of the culture with other components. This requires careful dosage of the powder (nutrients) and of all the other components to obtain an efficient

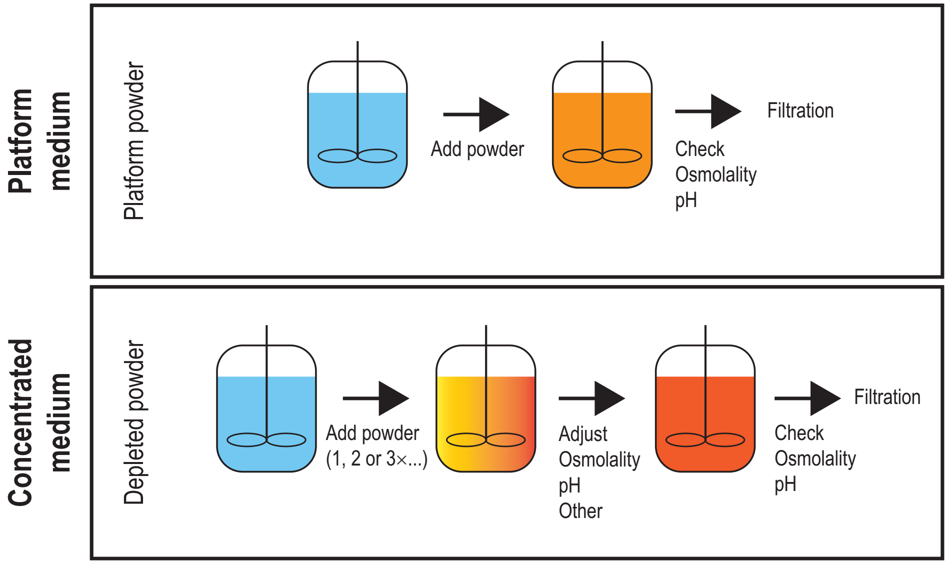

Figure 2.12 Example of media preparation. Production environment (top): the platform powder containing all medium compounds is solubilised, and after quality check (pH and osmolality), the medium is filtered and used. Process development environment (bottom): the depleted powder is added to the extent needed for a certain nutrient concentration, and then the formulation is adjusted (pH and osmolality) using baseline components before being filtered and used.

medium for the overall needs of the cell culture. In practice, this is typically done by properly combining a so-called basal medium, which corresponds to a formulation that is concentrated 1× and includes all the baseline compounds, with formulations containing only nutrients with higher concentrations, as discussed in the following example.

Let us consider, for example, the case where we are ideally operating the bioreactor at the $CSPR_{min}$ and at a certain VCD value, and we now want to increase the VCD twice. As discussed earlier, we can double the perfusion rate so as to keep constant the CSPR value and not drop it below its minimum value, which would lead to an unstable culture. An alternative option would be to concentrate the medium formulation to decrease the $CSPR_{min}$ of the given medium and cell line combination. A third option that would allow using a fixed basal medium would be the use of a secondary feed, highly concentrated in only the nutrients. The flowrate of this feed would be negligible or at least very low compared to that of the basal medium, but the overall medium depth, when considering the total perfusion rate given by the basal medium and the secondary feed, could be significantly increased. An example of such an application was given by Karst et al. (2016), although in that case, the concentrated feed was premixed to the basal medium. This approach provides a very flexible basis to change the medium depth while using well-defined and fixed streams: the basal medium and a concentrated nutrient secondary feed. Another advantage is that solubility or stability limitations faced by some components in the complex medium formulation can be reduced in the secondary feed. Solubility of amino acids, for example, varies with pH and may become limiting for the medium stability (McCoy et al. 2015). The pH of the feed can, in fact, be adjusted according

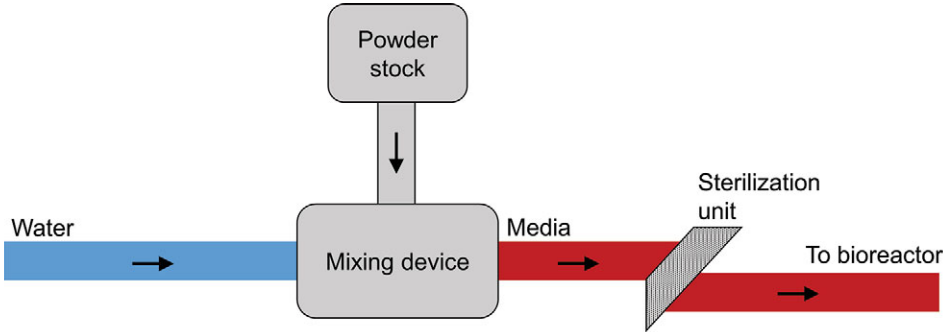

Figure 2.13 Schematic of an online media preparation device.

to solubility limitations without being critical since it is used in combination with the medium that has some pH buffer capacity.

Finally, it is worth to at least mention the possibility of preparing media by sterile online mixing – for example, from a single powder, as shown schematically in Figure 2.13. Using such a device, media could be produced on-demand and storage space would be reduced significantly. From a technical point of view, such a device has to fulfil several requirements, such as tight control of the homogeneity of the solution and of the added amount of powder. In addition, a careful characterisation of mixing times, pH adjustments and other media-specific constraints (solubility and buffer capacity) are necessary and potentially challenging to realise.

2.6 STEADY-STATE OPERATION AND PROCESS DYNAMICS

2.6.1 The Continuous Stirred Tank Reactor Model

The aforementioned equipment and control strategies enable the continuous operation of mammalian cell perfusion cultures at constant viable cell density and provide the basis for steady-state operation. Assuming uniform composition and temperature within the bioreactor, its behaviour can be described in the frame of the classical continuous stirred tank reactor (CSTR) model. With reference to a generic species i, the mass balance in the CSTR is given by the following equation:

$$V_R \frac{dt}{dc_i} = Qc_i^{in} - Qc_i + r_i V_R, \tag{2.5}$$

where V_R is the reactor working volume, assumed to be constant, Q is the volumetric flowrate; c_i is the concentration of the species i inside the reactor; $c_{i,in}$ is its concentration in the inlet stream; and r_i is the corresponding reaction production term. Dividing both terms by V_R and using Equations (2.1) and (2.2) allows to reformulate this equation in terms of the volume specific rates as follows:

$$\frac{dc_i}{dt} = Pc_i^{in} - (H + B)c_i + r_i, \tag{2.6}$$

where the term on the left-hand side represents accumulation inside the reactor, while the three terms on the right-hand side represent the contribution of inlet (perfusion), outlet (harvest and bleed) and production rate, respectively. A similar mass balance can be written for the viable cell density leading to

$$\frac{dX_v}{dt} = (\mu - B)X_v, \tag{2.7}$$

where the accumulation term is set equal to the difference between the cell growth rate, μ, and the bleed rate. The first one describes the cell-doubling process and is a function of the cell metabolism and consequently of the entire cell environment inside the reactor. Dividing Equations (2.6) and (2.7) by the viable cell density X_v allows calculating cell-specific accumulations:

$$\frac{dc_i}{dt}\frac{1}{X_v} = \frac{(Pc_i^{in} - (H+B)c_i)}{X_v} + q_i \tag{2.8}$$

$$\frac{dX_v}{dt}\frac{1}{X_v} = \mu - B, \tag{2.9}$$

where the cell-specific production rate of the species i, q_i, represents the ratio r_i/X_v and is related not only to chemical reactions but also to biological processes involving the cell metabolism.

By operating a CSTR with constant inlet flows, it is possible, after a certain transient period of time, to achieve a stationary state, where the inlet streams and production processes balance the outlet streams and consumption ones. In this state, all the process variables are invariant in time and the environment inside the reactor, which was already uniform in space, also becomes constant in time. This is the ideal situation for the cells to always follow the same path in expressing the target protein. Therefore, this leads to a product quality heterogeneity, in terms of glycoform and charge variant distributions, which is reduced to the minimum compatible with the living nature of the cells. In contrast, in a fed-batch process, the environment changes during the entire process, with depletion of the nutrients and accumulation of the toxic metabolites. This affects the cell metabolism, leading to differences in the target protein expressed during the run, and then increasing heterogeneity of the corresponding quality attributes. As a consequence, product quality attributes are much more heterogeneous in batch than in continuous operations.

At steady state, there is no accumulation of any species inside the reactor, and consequently, mass balances 2.8 and 2.9 reduce to

$$q_i = \frac{(H+B)c_i - Pc_i^{in}}{X_v} \tag{2.10}$$

$$\mu = B. \tag{2.11}$$

It is worth pointing out that the notion of steady state in living systems has to be taken with some care. In particular, the cell-specific productivity, q_i, and the cell growth rate, μ, are functions of the cell metabolism and therefore may be affected by several processes related to cell ageing or high cell generation numbers. In some cases, for example, this has been observed to lead to the decrease of the cell-specific productivity of a mAb over the culture time (Karst et al. 2017a, Ozturk 2014). In general, these processes are, however, rather slow and occur over several days. In particular, they are much slower than the bioreactor characteristic time, which is related to the average residence time and is in the order of one day. This implies that the bioreactor dynamics is usually fast enough to adapt to the slow changes of q_i and μ, thus establishing so-called pseudo-steady-state conditions. This is an important general concept in system dynamics, which means that the accumulation terms, although non-zero, are negligible with respect to the other terms in the mass balances, so Equations (2.10) and (2.11) still hold true, with all the involved terms changing slightly in time. In other words, the input and output terms in the mass balances compensate each other instantaneously (the reactor dynamics) although they all change on a long time scale (the cell metabolism dynamics).

2.6.2 Residence Time Distribution

In a perfusion bioreactor, fresh and spent medium are continuously exchanged. The time that every single molecule spends inside the bioreactor system is described by the residence time distribution (RTD). The average residence time, τ, is given by the ratio between the reactor volume, V_R, and the volumetric exchange rate, Q, which is equal to the inverse of the normalised perfusion rate:

$$\tau = \frac{V_R}{Q} = \frac{1}{P}. \tag{2.12}$$

As mentioned earlier, a perfusion bioreactor can be described through the CSTR model, and therefore, the corresponding residence time distribution is given by the typical exponentially decreasing function:

$$\text{RTD} = E(t) = \frac{1}{\tau} \times exp^{-\frac{t}{\tau}}. \tag{2.13}$$

The preceding equations allow quantifying the influence of the perfusion rate on the residence time of all molecules inside the reactor. The higher the perfusion rate, the shorter the average residence time value and the sharper its distribution. For example, for medium perfusion rates going from 0.5 to 2 RV/day, the fraction of molecules that stays in the reactor less than one day increases from 39 to 86 per cent, as shown in Figure 2.14A and B. By comparison, in a classical fed-batch process lasting 14 days, the RTD looks quite different, as shown in Figure 2.14C. The protein is produced over the entire run, and, consequently, each single molecule has a different exposure, ranging from zero to 14 days, to post-translational modifications. This leads to very heterogeneous product quality attributes, since proteins produced in early stages are more likely to be modified than proteins produced at later stages. This, in addition to the effect of the changing cell environment discussed earlier, further contributes to the fact that product quality attributes are much more heterogeneous in batch cultures than in continuous cultures.

2.6.3 Steady-State Characterisation

Due to the presence of living cells, the processes taking place in a bioreactor are far more complex than those occurring in a typical chemical reactor. This also reflects on the definition of steady state and its characterisation. In particular, we need to distinguish between extracellular (e.g., glucose, lactate and amino acids) and intracellular processes which involve different chemical and biological entities and, therefore, have to be analysed separately. The chemical composition of the supernatant can be easily analysed, and it has been shown to become constant in time once steady state is achieved after a proper transient time, resulting in constant cell-specific production and consumption rates of the involved metabolites (Karst et al. 2017a). A detailed analysis based on metabolomics (Karst et al. 2017c, 2017d) has demonstrated that the concentrations of intracellular metabolites – such as nucleotides, nucleotide sugars and lipid precursors – also reach constant levels in a perfusion culture at constant cell density and perfusion rate. By analysing the cellular transcriptome and proteome, it has been confirmed that most of the identified transcripts and intracellular proteins fulfil the steady-state conditions (Bertrand et al. 2019). For a minority of transcripts, no steady state was achieved. Although no correspondence with a specific biological path could be identified, this may indicate that even if steady-state conditions can be achieved in perfusion operation, both intra- and extracellular, some biological events that occur with minor macroscopic effects may still be varying and changing over the time course of the operation.

Due to the complexity of these processes and the intrinsic variability of living systems, the operation of mammalian cell perfusion cultures at constant cell density

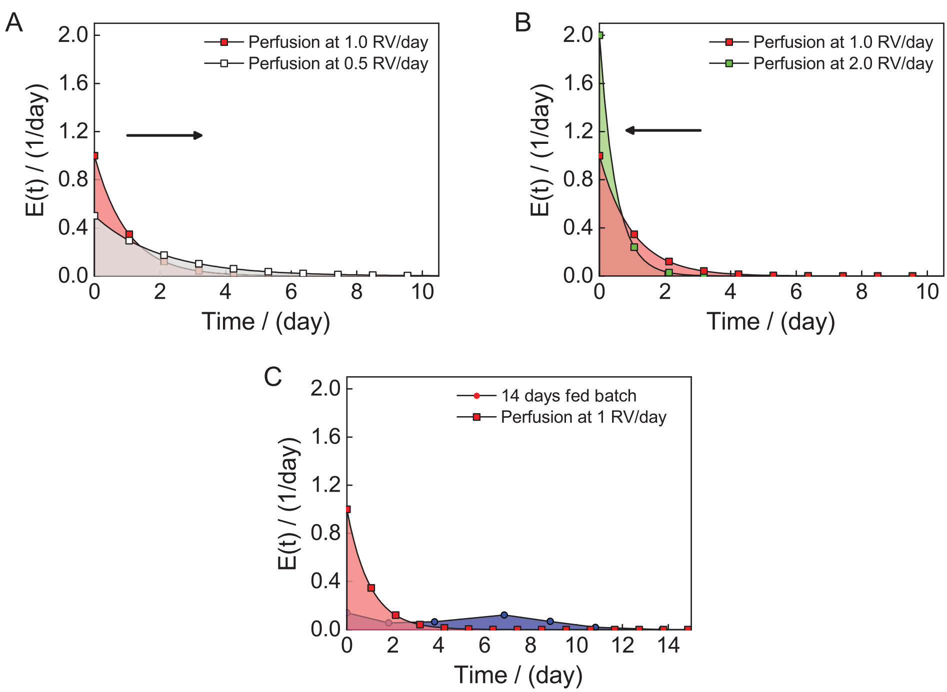

Figure 2.14 Comparison of the residence time distribution of the target protein in perfusion bioreactors at different perfusion rates and in a fed-batch bioreactor: (A) Decreasing the perfusion rate from 1.0 to 0.5 RV/day in a perfusion bioreactor leads to a broader RTD, (B) increasing the perfusion rate from 1.0 to 2.0 RV/day sharpens the RTD, and (C) in a 14-day fed-batch the RTD is very wide with an average value of 7 days, while more than 60 per cent of the produced proteins spend less than one day in a perfusion system operated at a perfusion rate of 1 RV/day.

and perfusion rate is sometimes referred to as state of control instead of steady-state operation (Nasr et al. 2017). This means critical process parameters and product quality attributes are kept in a well-defined narrow range of values with only small perturbations, rather than at a fixed value. Appropriate control strategies enable operation at this state, which allows the production of the target protein with sufficiently uniform quality.

2.6.4 Transient to Steady State

In a perfusion bioreactor, various processes characterised by different dynamics take place simultaneously, and their superimposition determines the dynamic behaviour of the whole unit. These include, on the one hand, the bioreactor hydrodynamics that is mainly defined by the medium exchange rate and, on the other hand, the cellular dynamics (affecting growth, consumption and production rates) defined by the cell metabolism.

The hydrodynamics of the reactor is expected to reach steady state faster, as the cellular response in a constant environment might lag behind. The reference point day zero is defined as the time where the VCD set point is reached. For a perfusion rate of 1 RV/day, it has been observed that extracellular metabolites take up to three days in order to achieve constant levels. This corresponds to three times the average residence time, τ, which is known to be the typical time for reaching steady state in a continuous stirred vessel. Similar trends have been observed for cell-specific rates as the growth rate and glucose uptake or lactate consumption rates, underlining that the cellular exchange rates with the environment follow the reactor hydrodynamics (Karst et al. 2016). With respect to intracellular CHO cell metabolites, Karst et al. (2017c) observed similar trends for the energetic state of the cell, as the energy charge, the uridine (U) and the nucleotide triphosphate (NTP) ratio achieved constant levels about three days after a set point change.

The detailed analysis of the intracellular proteome and transcriptome has revealed three major dynamics in a perfusion bioreactor operated at 1 RV/day, each corresponding to a different set of chemical and biological entities: one is completed and reaches steady state after about three days, another one after seven days and the last one never reaches a time constant state or, if it does, only after extremely long times. The first two are illustrated in Figure 2.15, taking as day zero the one where constant VCD is achieved. It is seen that the hydrodynamic transient of the reactor, and the corresponding extra- and intracellular metabolites mentioned earlier, is over in about 3τ. On the other hand, some of the product quality attributes exhibit a longer dynamics. In particular, patterns of the N-linked glycans took between five and seven days to reach stable levels, while charge isoform patterns took three to four days (Karst et al. 2017a, 2018; Walther et al. 2018). These results indicate that the reactor hydrodynamics dominate the dynamics of most of the process variables. However, some intracellular processes, like those involved in the synthesis of glycoforms, lag behind and take longer.

The longest observed dynamics is related to the intrinsic nature of living systems. As mentioned earlier in the context of Equations (2.10) and (2.11), instabilities of the cell lines due to genetic or epigenetic changes, while going through multiple generations or doublings, might lead to a decrease in cell-specific productivity. For example, Karst et al. (2016) observed a 30 per cent drop in productivity over 90 generations,

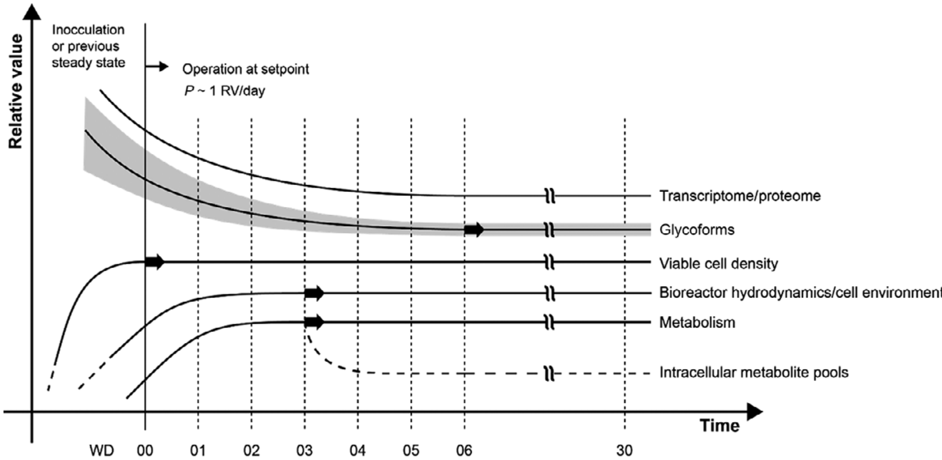

Figure 2.15 Different dynamics in a perfusion bioreactor reaching steady-state conditions. Constant viable cell density defines the reference day. Metabolism and bioreactor hydrodynamics take about three average residence times – i.e., three to four days, while product quality takes about six to seven days to reach steady state.

and even larger changes are reported by Ozturk (2014) during the entire process run, without never achieving steady-state conditions. However, these are typically very slow dynamics which, per se, do not cause significant difficulties with respect to the reactor operation, which can then be regarded as a pseudo-steady-state, as discussed earlier. On the other hand, in the long term, these processes reduce the bioreactor productivity and possibly also the product quality and, therefore, cannot be tolerated. This highlights the importance of developing robust and stable cell lines specifically dedicated to perfusion processes.

2.6.5 Product Quality in Batch and Continuous Reactors

Whether we refer to steady state or state of control, mammalian cell perfusion cultures clearly benefit from the reactor environment which is uniform in space and constant in time, as well as from the short and narrow residence time distribution of the target product. This results in a more uniform cell environment which promotes low product heterogeneity and shorter times for post-translational modifications. In addition, the continuous exchange of medium prevents the accumulation of inhibitory or toxic by-products, like lactate or ammonium. In contrast, in a fed-batch culture, the environment changes during the entire reactor run so that the cells experience a continuously changing environment, thus promoting corresponding changes in the product quality attributes, such as the patterns of N-linked glycosylation and charge variants. In addition, the broad RTD of the target protein contributes to this heterogeneity, leading to different modification degrees, depending on the time spent by the single protein inside the reactor after its expression. With illustrative purposes, in Figure 2.16, the charge variant distribution obtained in the same system but operated in the fed-batch and in the continuous mode are compared. It is seen that – as expected, based on the previously described processes – the distribution obtained in the continuous mode is narrower, thus indicating a less heterogeneous target protein (Karst et al. 2018).

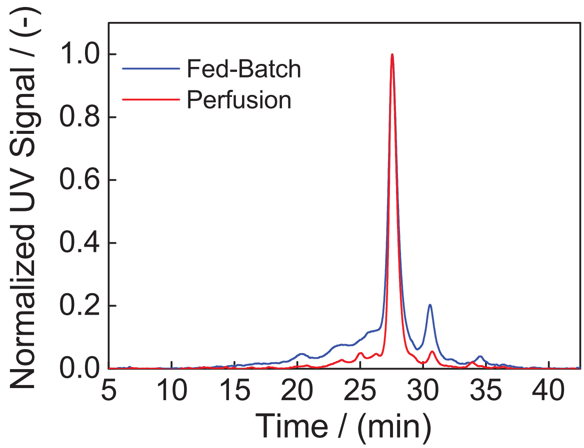

Figure 2.16 Comparison of the chromatograms of charge isoform distribution for a commercial mAb produced in fed-batch and in perfusion mode. © Reprinted from *Current Opinion in Biotechnology*, 53, Daniel J. Karst, Fabian Steinebach and Massimo Morbidelli, 'Continuous integrated manufacturing of therapeutic proteins', 2018, with permission from Elsevier

2.7 CONCLUSION

In this chapter, we have discussed the fundamentals of operation and control of perfusion bioreactors. Although these concepts are valid in general across scales, our reference was the bench-scale bioreactor at the litre scale. We have introduced the different equipment needed to realise a perfusion process and have indicated the critical aspects in the operation of each one of them. The most important operating and performance parameters have been considered, with particular reference to the CSPR, which is peculiar of the perfusion operation. Another important aspect in these systems is the media design, which needs not only to guarantee a sufficient supply of nutrients but also to keep physiological conditions during the entire operation – in particular, in terms of osmolality and pH. Although the currently used media are not specifically designed for perfusion operation, this can achieve large productivities based on high cell density and elimination of waste metabolites.

Perfusion bioreactors exhibit a distinctly different behaviour compared to batch or fed-batch units. This, in general, includes an initial transient behaviour followed by steady-state conditions which are ideal to produce consistently high quality and homogeneous products. In addition, continuous operation allows the achievement of very small residence times of the product inside the reactor, which limit all the following processes that affect the product quality, such as decomposition reactions or aggregation processes.